EVPK Simulator

User Manual

A practical guide to the web app at igem.ardabakici.com for pharmacokinetic (PK) modeling (1c/2c/3c), regimen comparison, and pharmacodynamic (PD) links (bacteria CFU kill and PMM2 activity rescue). Screenshots and button-by-button instructions are included.

Source code: github.com/Aman-Sunesh/EV-PK-Simulator

October 6, 2025

Contents

1 Introduction

The PB–PK Simulator is a browser-based toolkit for:

- Fitting 1-, 2-, or 3-compartment macro-exponential PK models to CSVs.
- Simulating dosing routes and regimens (IV bolus, short infusion, oral; repeat dosing).
- Comparing multiple regimens on one plot with key KPIs.
- Performing PD analyses driven by PK outputs: bacteria CFU dynamics and PMM2 activity rescue.
- Generating one-click PDF reports that capture parameters, selection/demotion rationale, plots, and data-hygiene warnings.

Direct access. The hosted app is available at igem.ardabakici.com.

Run locally (optional). If you prefer a local instance, follow the backend/frontend steps in Section 2.

2 Installation (Local Development)

2.1 Prerequisites

Python 3.10+ and Node.js 18+ recommended.

2.2 Backend

```
# From repo root
pip install -r requirements.txt

cd backend
python3 -m venv venv
source venv/bin/activate

# Run FastAPI dev server on http://0.0.0.0:8000
uvicorn main:app --reload --host 0.0.0.0 --port 8000
```

2.3 Frontend

```
cd frontend
npm install
npm start
```

The frontend dev server typically opens at `http://localhost:3000` and proxies API calls to the backend (`http://localhost:8000`).

3 App Overview (Home)

The home page presents the main modules as large tiles: One-/Two-/Three-Compartment Models, PD Analysis from Data, and Compare Regimens.

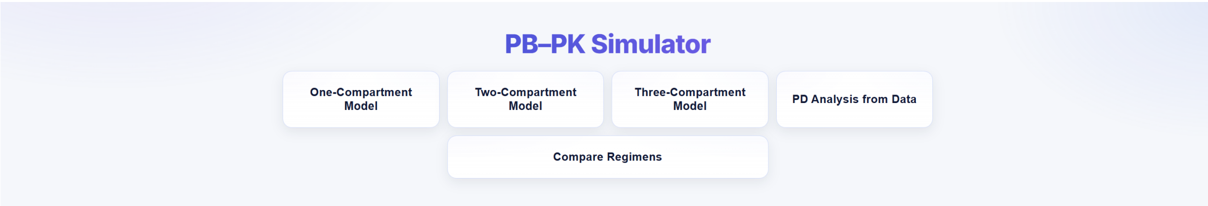

Figure 1: Home gateway tiles to PK and PD tools.

Click a tile to open that module. Navigation links at the top-left return you to the Home at any time.

4 Three-Compartment Model (PK)

This screen is representative of the single- and two-compartment pages; fields differ only by the number of macro terms.

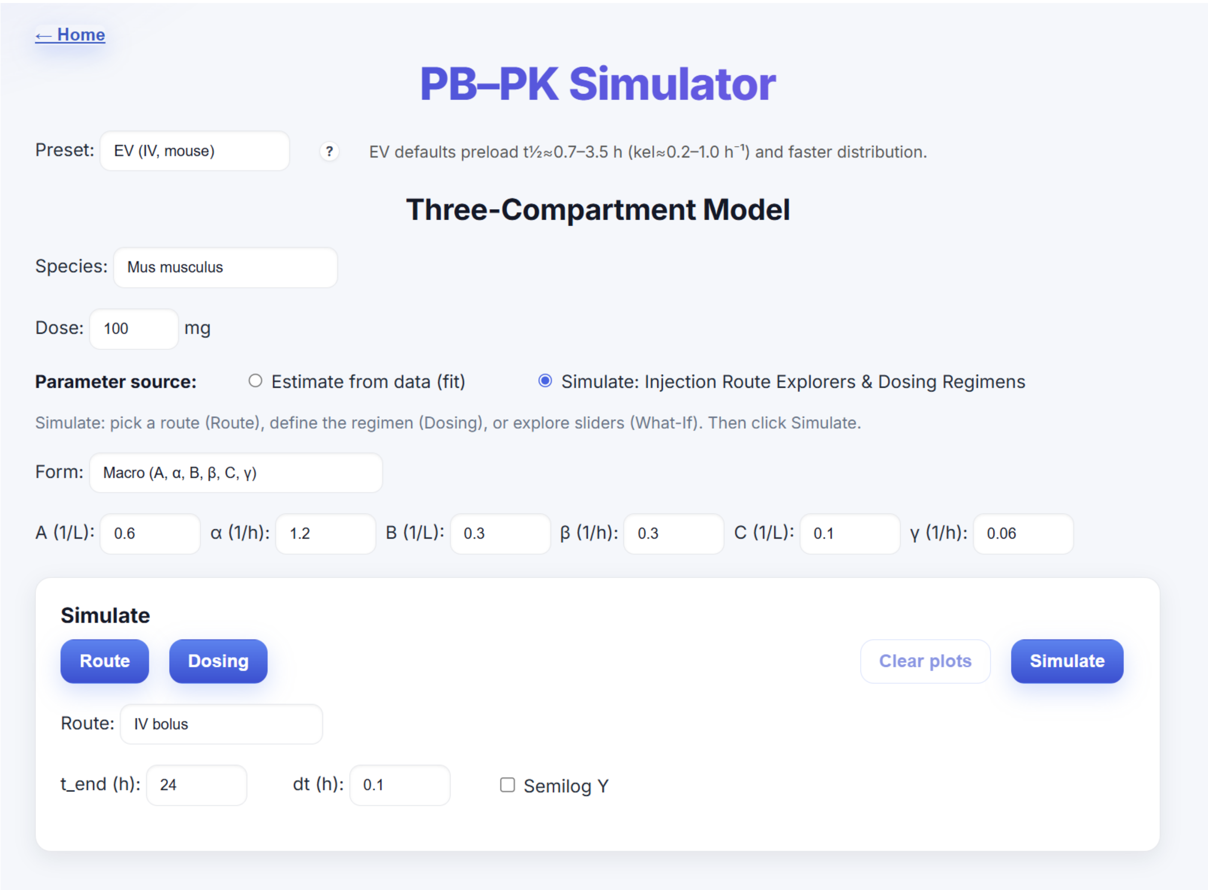

Figure 2: Three-Compartment model — parameter inputs and simulate panel.

4.1 Header & Preset

- **Preset**: quick-fill convenience values (e.g., species-specific defaults).
- Inline tip text reminds IV bolus half-life ranges and distribution expectations.

4.2 Inputs

- **Species**: free text (for labeling).
- **Dose (mg)**: total amount for a single administration.
- **Parameter source**:
  - *Estimate from data (fit)* — upload a CSV to fit macro parameters.
  - *Simulate: Injection Route Explorers & Dosing Regimens* — directly enter macro parameters and simulate.
- **Form**: macro parameters. For 3c you enter $A, \alpha, B, \beta, C, \gamma$; for 2c omit (C, γ); for 1c you can use (V_d, k) or (A, α) form.

4.3 Simulate Panel (bottom of page)

- **Route**: choose the input route (IV bolus; IV infusion (short, overlap/non-overlap); oral with first-order absorption).
- **Dosing**: define dose events: bolus at time t, infusion with start/duration, oral repeats, or a programmatic repeat every τ.
- t_{end} **(h)**: simulation horizon. dt **(h)**: time step.
- **Semilog Y**: toggles a log-scale concentration axis.
- **Buttons**: **Clear plots** (reset) and **Simulate** (run and plot).

Output. The plot shows $C(t)$; above it KPIs such as **Cmax**, **Tmax**, and **AUC (0–t_{end})** are summarized.

5 Simulate: Route/Dosing Builder (All Models)

This screen exposes a plan-builder when you click **Route** or **Dosing** (or open the dedicated *Simulate* page). You can stack multiple inputs and remove them.

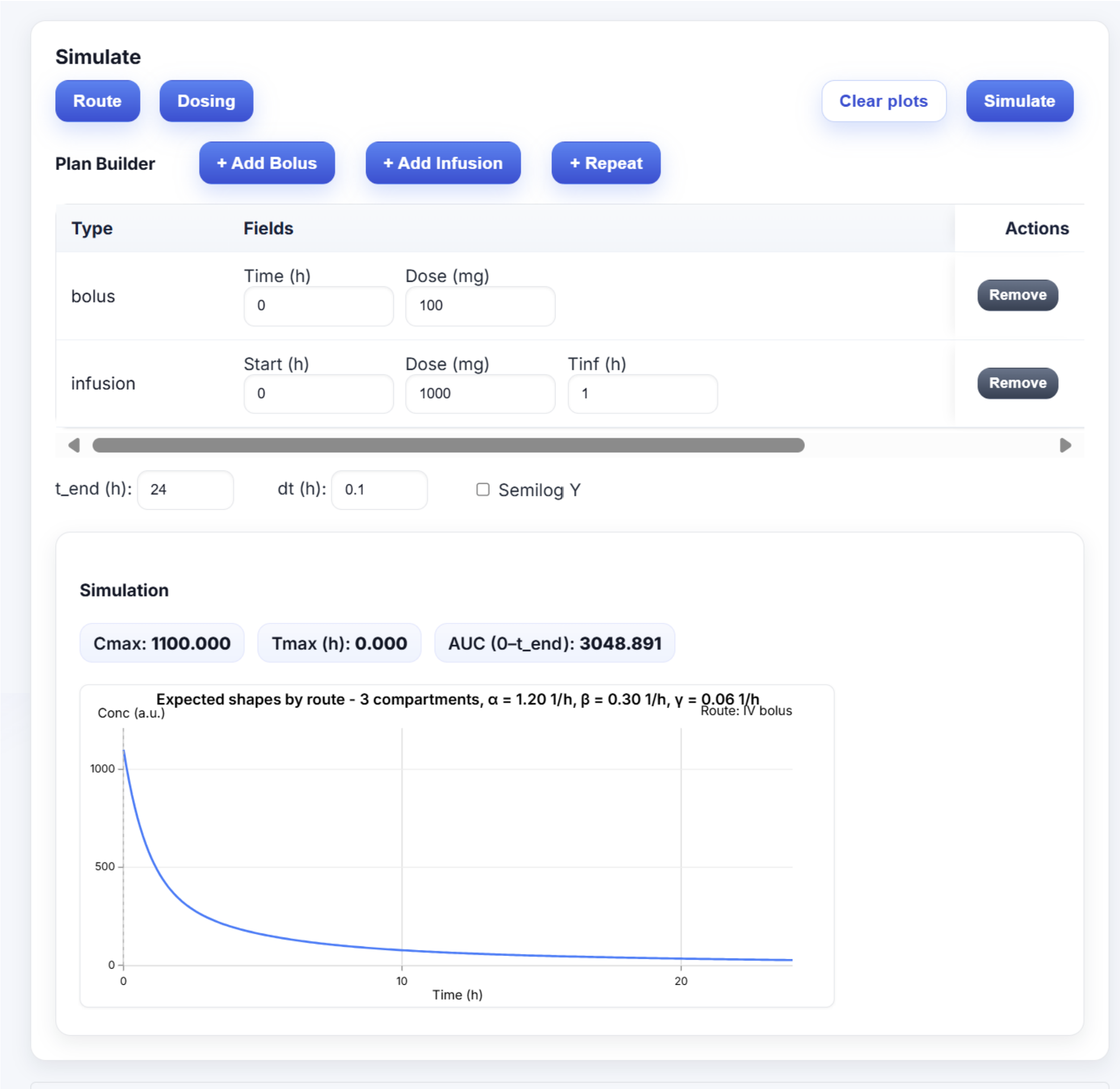

Figure 3: Plan Builder: add bolus & infusion events, then simulate. KPIs appear above the plot.

Controls:

- **+ Add Bolus**: Time (h), Dose (mg). Creates an impulse input.
- **+ Add Infusion**: Start (h), Dose (mg), T_{inf} (h). Adds a zero-order input lasting T_{inf}.
- **+ Repeat**: repeat a prior pattern every τ hours (steady-state experiments).
- **Actions** $\rightarrow$**Remove**: delete a row.
- Bottom controls: t_{end}, dt, Semilog Y toggle, and **Simulate**.

Readouts:

- **Cmax**, **Tmax**, **AUC**: auto-computed for the shown window.
- Plot: linear (or semi-log) concentration profile.

6 Compare Regimens (Side-by-side PK)

Define multiple scenarios across rows, then render them on a common timeline.

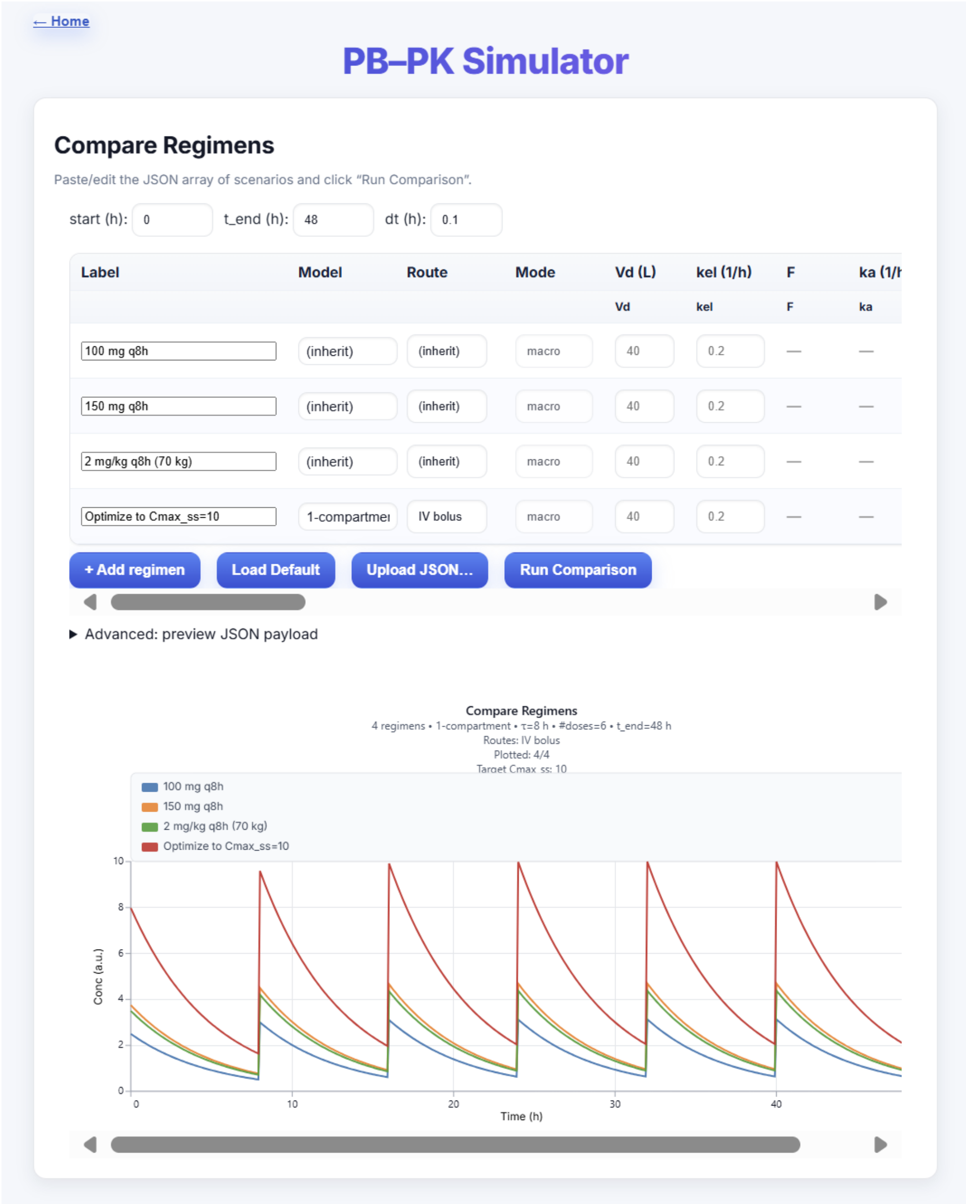

Figure 4: Compare up to many regimens with a shared time base. Advanced: preview JSON payload.

Top timeline fields (global to the comparison):

- **start (h)**, **t_end (h)**, **dt (h)**: simulation window and resolution.

Regimen table (one row per scenario):

- **Label**: legend name.
- **Model**: inherit (use the current page model) or override.
- **Route/Mode**: IV bolus or infusion; macro vs micro entry (usually macro).
- **Vd**, **kel**, **F**, **ka**: visible when relevant for 1c macro/micro forms.

Buttons:

- **+ Add regimen**: append a new row.
- **Load Default**: populate a few example rows.
- **Upload JSON...**: load a saved comparison payload.
- **Run Comparison**: compute and render.

Advanced (accordion):

- *Preview JSON payload*: inspect the exact request that the UI sends.

Output: Overlaid $C(t)$ curves with legend and optional target annotations (e.g., *Target Cmax,ss = 10*).

7 PD Analysis from Data

This module accepts raw *concentration–time* CSVs (from your lab or a PK simulation) and computes PD outcomes using built-in link models.

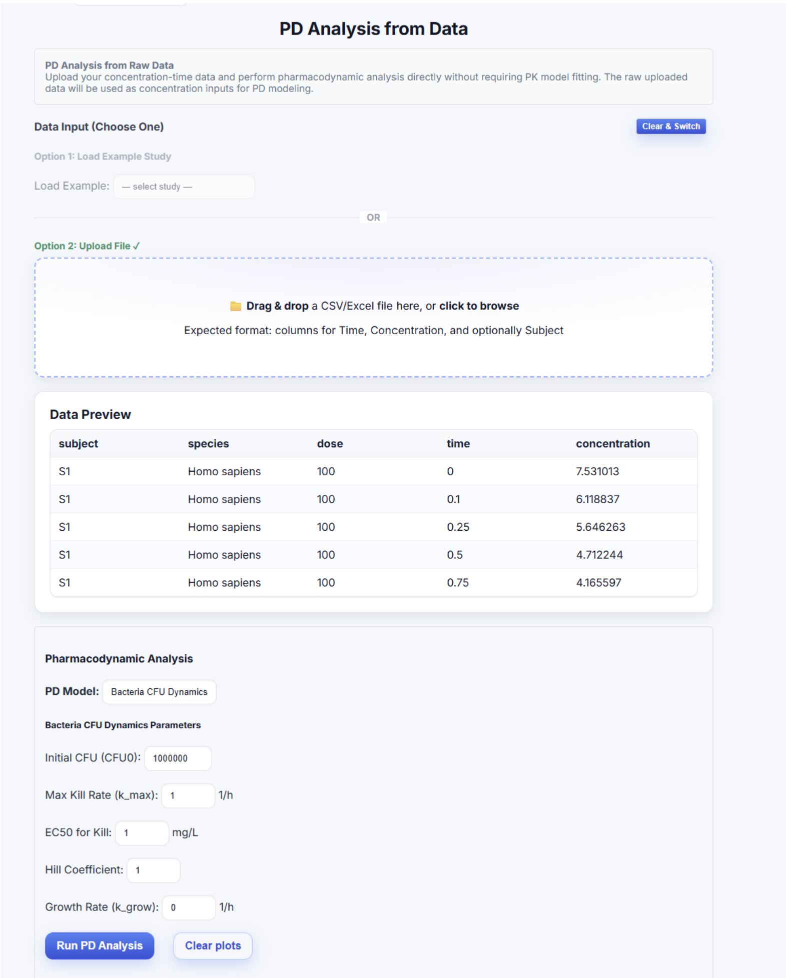

Figure 5: PD Analysis from Data: drag & drop CSVs, preview the table, then configure the PD model.

7.1 Data Input

- **Load Example**: pick a built-in small dataset (sanity check).
- **Drag & drop / click to browse**: upload your CSV/Excel. Expected columns include time, concentration, and optionally subject. The preview table displays the parsed data.

7.2 Pharmacodynamic Analysis Panel

- **PD Model** selector:
  - *Bacteria CFU Dynamics* (logistic growth with concentration-dependent kill).
  - *PMM2 Activity Rescue* (E_{max}/Hill link).
- **Buttons: Run PD Analysis**, **Clear plots**.

Outputs (after a run):

- Summary badges (e.g., *Initial CFU*, *Final CFU*, *Log Kill*).
- Time-series plots (e.g., $\log_{10}$ CFU vs time; %Activity vs time).
- **Download PD Report**: export a PDF with inputs, fitted/used PD parameters, and plots.

8 Standalone PD Workbench (Quick Run)

For quick bacterial-kill or PMM2-rescue explorations, you can open the PD workbench directly and enter PD parameters (without uploading data).

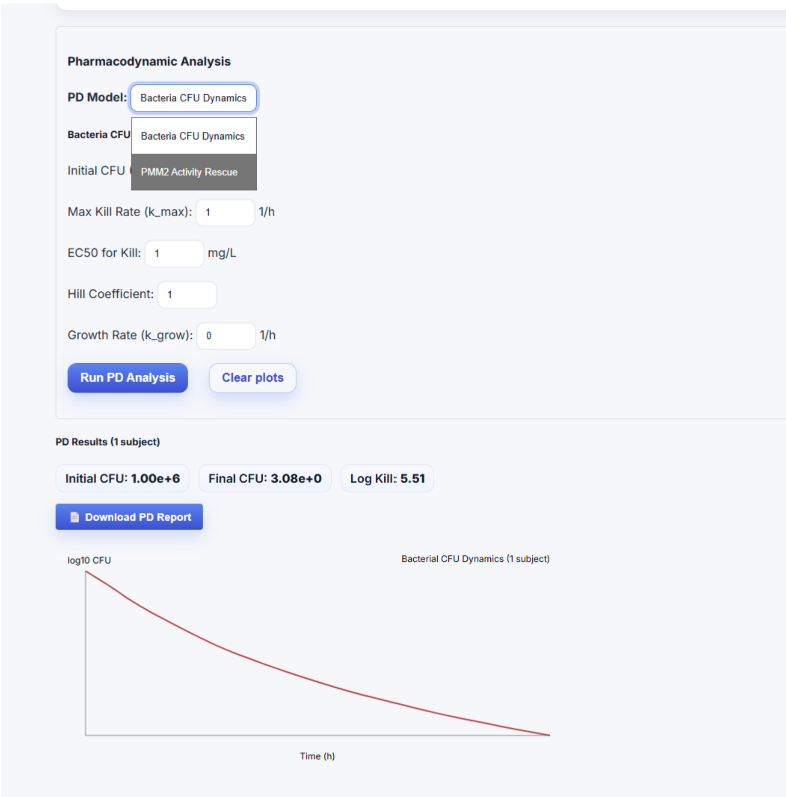

Figure 6: Bacteria CFU Dynamics: enter N_0, k_{max}, EC50, Hill h, and growth rate; then run.

8.1 Bacteria CFU Dynamics

Inputs:

- **Initial CFU** (CFU_0).
- **Max Kill Rate** (**$k_{\max}$**) [1/h].
- **EC50 for Kill** [mg/L].
- **Hill Coefficient** h.
- **Growth Rate** (**k_{grow}**) [1/h].

  Outputs:

- *Initial CFU*, *Final CFU*, *Log Kill*.
- Plot of $\log_{10}$ CFU vs time.
- **Download PD Report**.

8.2 PMM2 Activity Rescue

Inputs (typical):

- E_{basal}, $E_{\max}$, **EC50**, **Hill** h.

Output: %Activity time-course, suitable for target-attainment checks (e.g., maintain $\geq \theta\%$).

9 Fit From Data (PK)

When the **Parameter source** is set to *Estimate from data (fit)*, upload a CSV on the PK page to estimate macro parameters per subject. The *Data Input* screen (for example on the 3c page) looks like this:

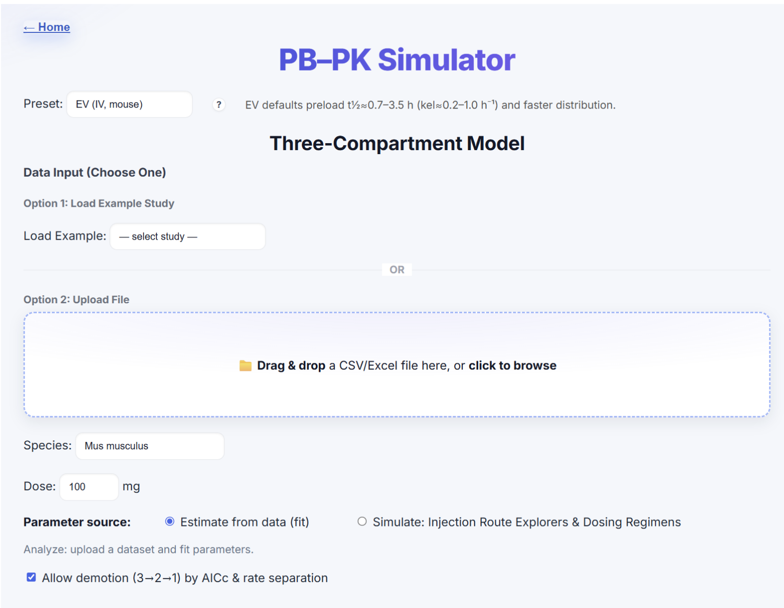

Figure 7: PK fitting data input: choose example or upload; species/dose are for labeling and scaling.

9.1 Model Selection & Demotion (under the hood)

The app fits 1c/2c/3c macro-exponential models and compares them with AICc. Identifiability guards demote 3→2 or 2→1 if:

- ΔAICc favors the simpler model (e.g., > 2),
- the minimum relative rate separation $\min_{i<j} \frac{|r_i - r_j|}{\max(r_i, r_j)}$ is too small (merged phases),
- the slowest tail AUC fraction is below a threshold (negligible tail).

Badges on the report explain the choice (selected model, ΔAICc, n, demotion reason, and any preprocessing warnings).

10 Mathematical Notes (Macro Forms)

For reference, single-dose closed forms used throughout (superposition handles repeats; steady state uses geometric series):

One compartment (1c).

$$C(t) = \frac{D}{V_d}\, e^{-k\,\Delta t}, \qquad \Delta t \equiv t - t_D \geq 0.$$

Two compartments (2c).

$$C(t) = D\left(Ae^{-\alpha\,\Delta t} + Be^{-\beta\,\Delta t}\right), \quad \alpha \neq \beta.$$

Three compartments (3c).

$$C(t) = D\left(Ae^{-\alpha\,\Delta t} + Be^{-\beta\,\Delta t} + Ce^{-\gamma\,\Delta t}\right), \quad \alpha > \beta > \gamma > 0.$$

Selection metric (linear-space residuals).

$$\mathrm{AICc} = 2k - 2\ln\hat{L} + \frac{2k(k+1)}{n-k-1}, \qquad \ln\hat{L} = -\frac{1}{2\sigma^2}\sum_i (y_i - \hat{y}_i)^2 + \text{const}.$$

Identifiability scores.

$$\mathrm{sep}_{\mathrm{rel}} = \min_{i<j} \frac{|r_i - r_j|}{\max(r_i, r_j)}, \quad r \in \{\alpha, \beta, \gamma\}, \qquad \phi_{\mathrm{tail,3c}} = \frac{C/\gamma}{A/\alpha + B/\beta + C/\gamma}.$$

11 CSV Schema, Units, & Hygiene

- Core columns (typical): `species`, `subject`, `time`, `concentration`, `dose`.
- Optional per-row dose: `dose_mg_per_kg`, `weight_kg` (enables subject-specific dosing).
- Units: time in hours; concentration in mg/L; dose in mg (unless stated).

Preprocessing policy:

- Drop non-positive concentrations (warning shown).
- Mean-aggregate duplicate `(subject, time)` rows (warning).
- Apply a tiny floor only for log plots to avoid log(0); outliers are otherwise kept.

12 Exporting Results

Most pages include **Download PDF report**. Exports typically contain:

- Parameter tables (with confidence intervals when available).
- Plots: linear/semi-log $C(t)$, residuals, or PD time-series.
- Badges: n, AICc/ΔAICc, selected model, demotion reason, and hygiene warnings.

13 Troubleshooting

Frontend

- **Blank plot/controls unresponsive**: refresh; check browser console for CORS errors if running locally.
- **"npm start" fails**: ensure Node 18+; remove `node_modules` & `package-lock.json` then `npm install`.

Backend

- **Port 8000 in use**: add `--port 8001` and update frontend proxy.
- **CSV rejected**: confirm required columns (columns "time" and "concentration" are required); ensure decimal points (not commas).

PK/PD specifics

- **Model demoted unexpectedly**: examine rate separation and tail-AUC badges; noisy data may not justify extra phases.
- **Noisy or sparse data**: prefer AICc-driven parsimony; try the Compare Regimens or Simulate route to explore design space.
- **PD run flatlines**: verify non-zero $k_{\max}$, reasonable EC50 and Hill; check units of input concentration.

14 Figures (Gallery)

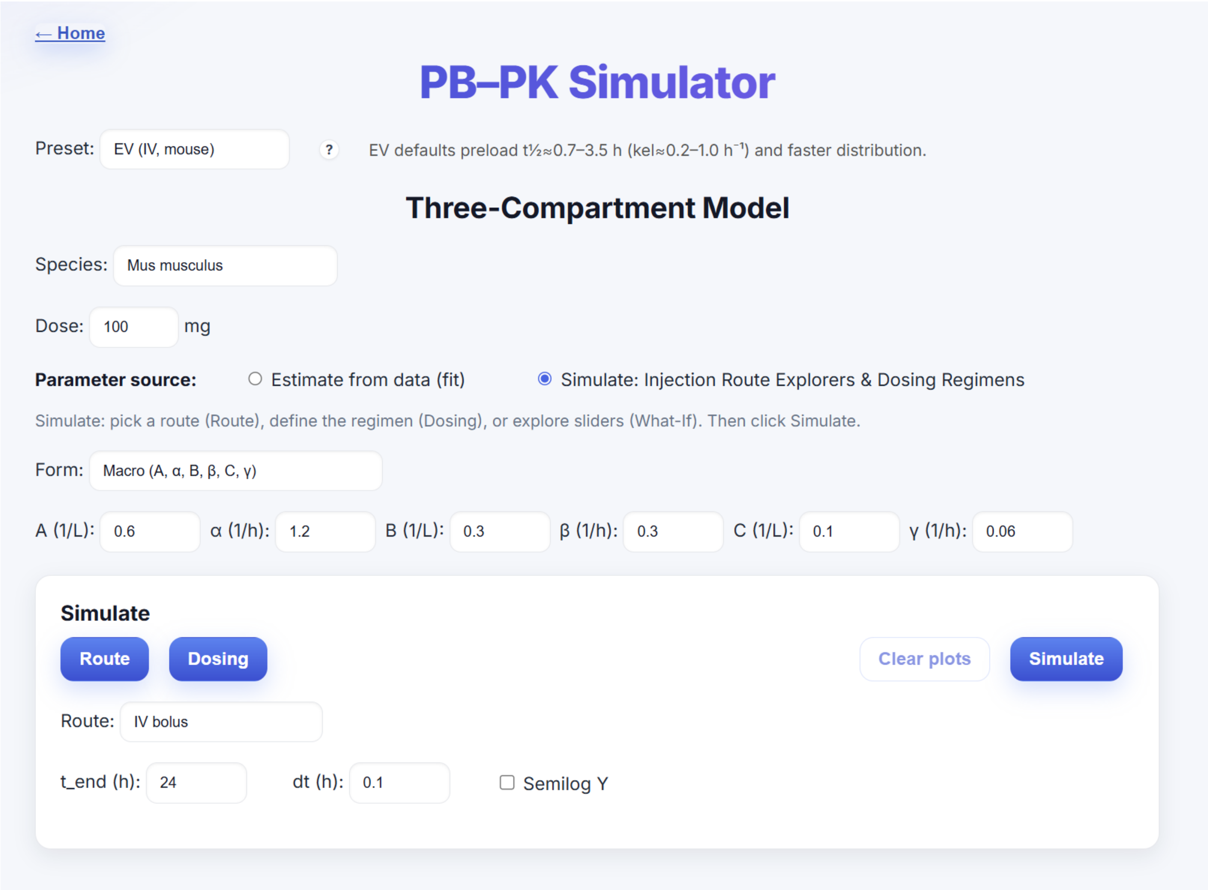

Figure 8: Three-Compartment Model (top).

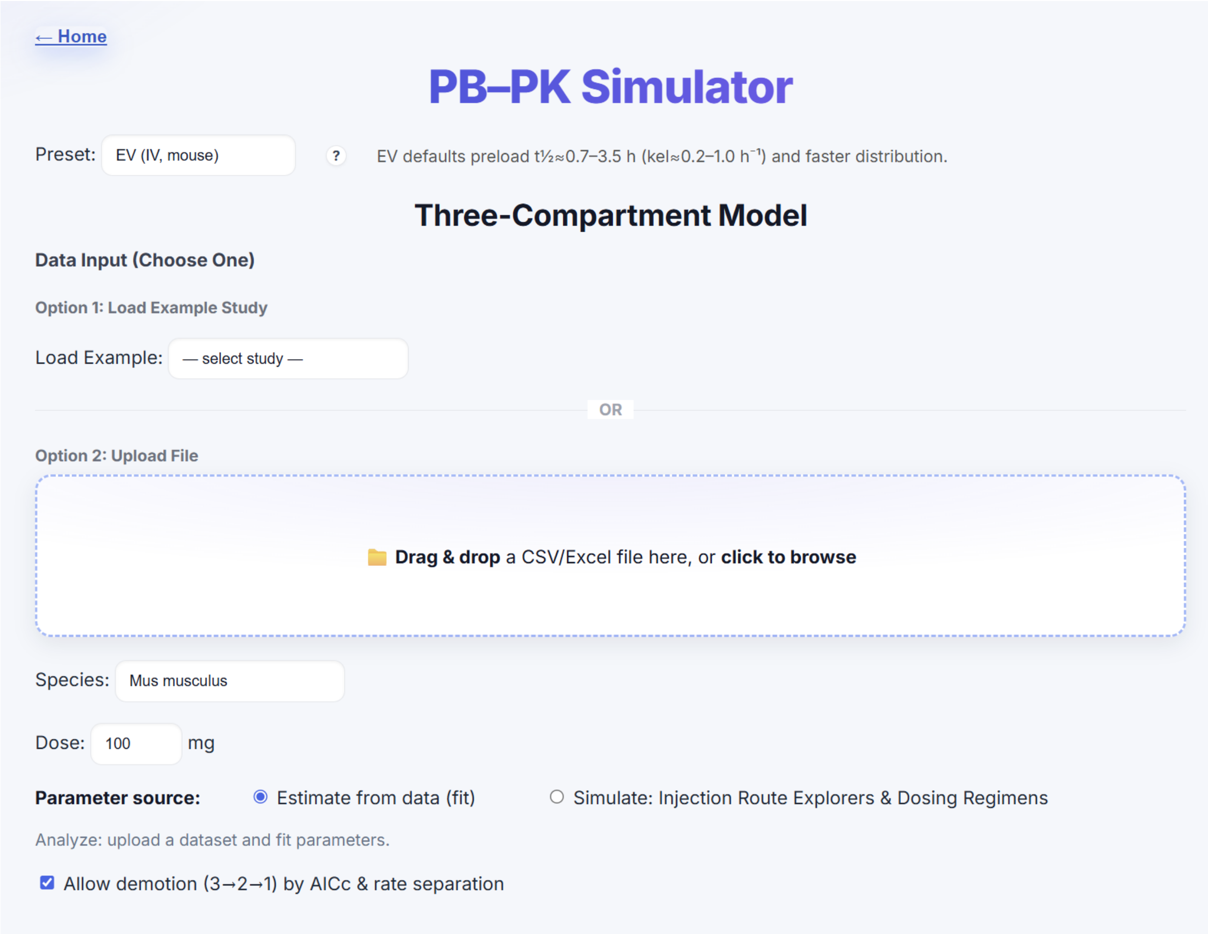

Figure 9: PK Fit: Data input & options.

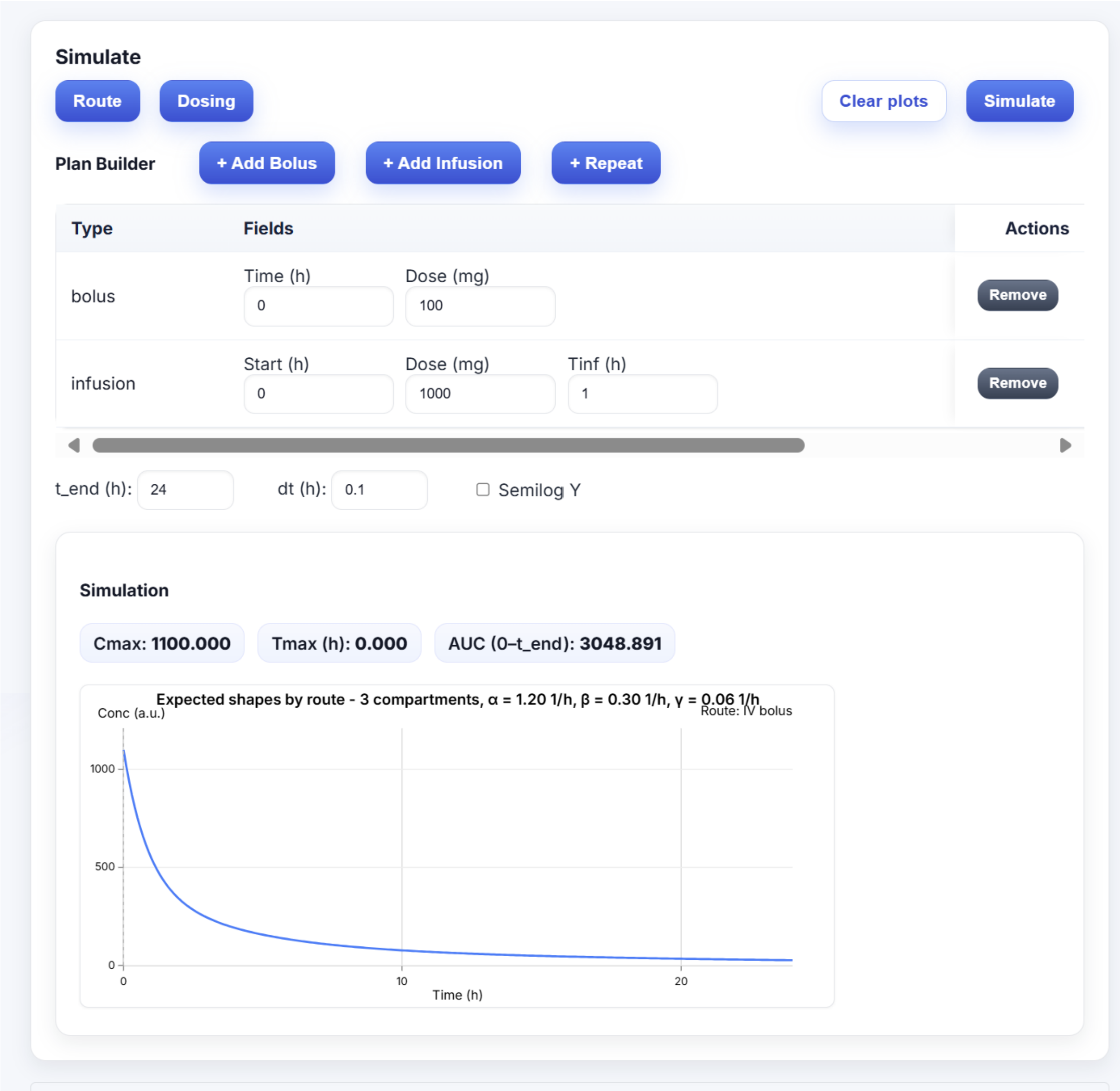

Figure 10: Simulate: Plan Builder and KPIs.

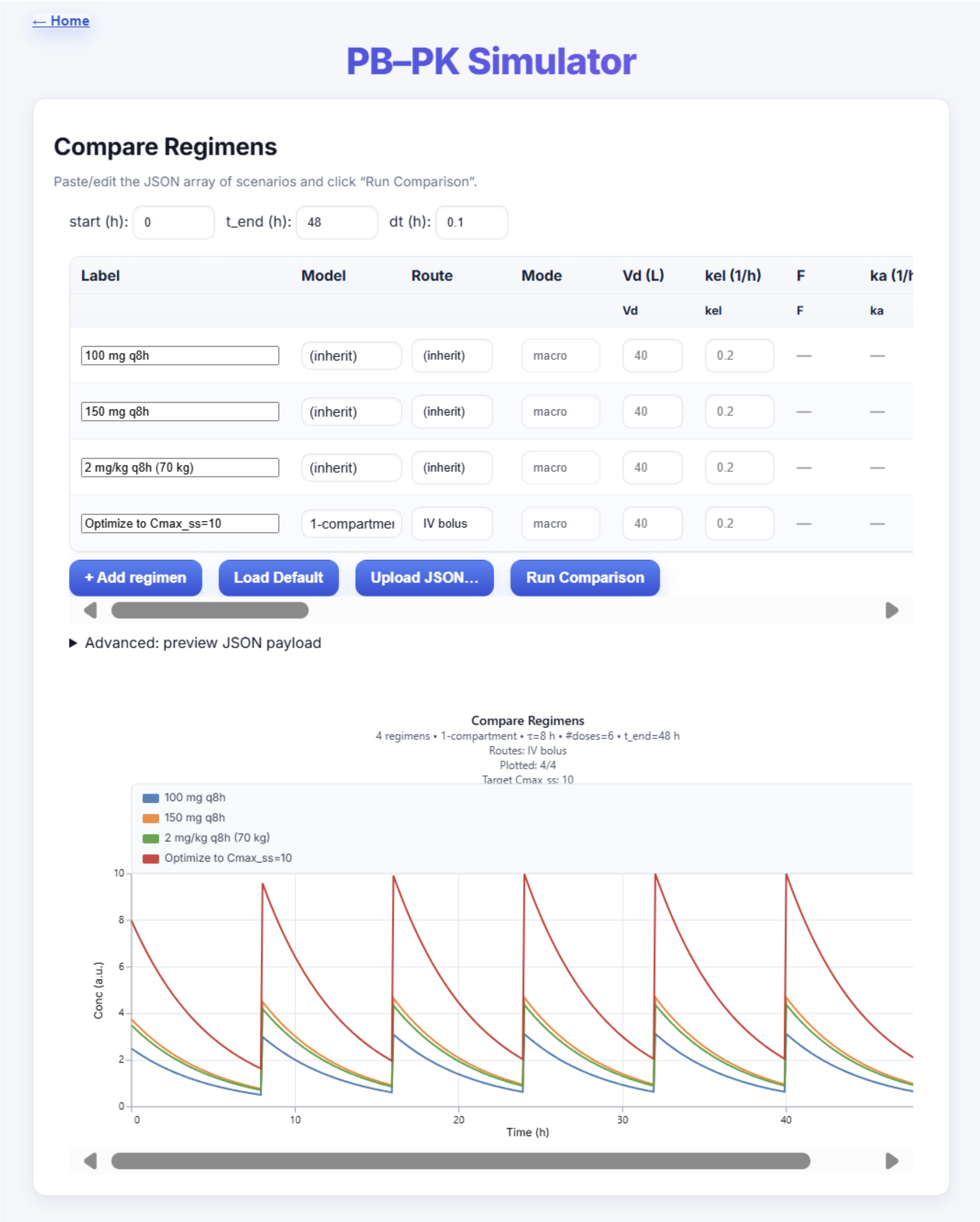

Figure 11: Compare Regimens: multiple scenarios on one plot.

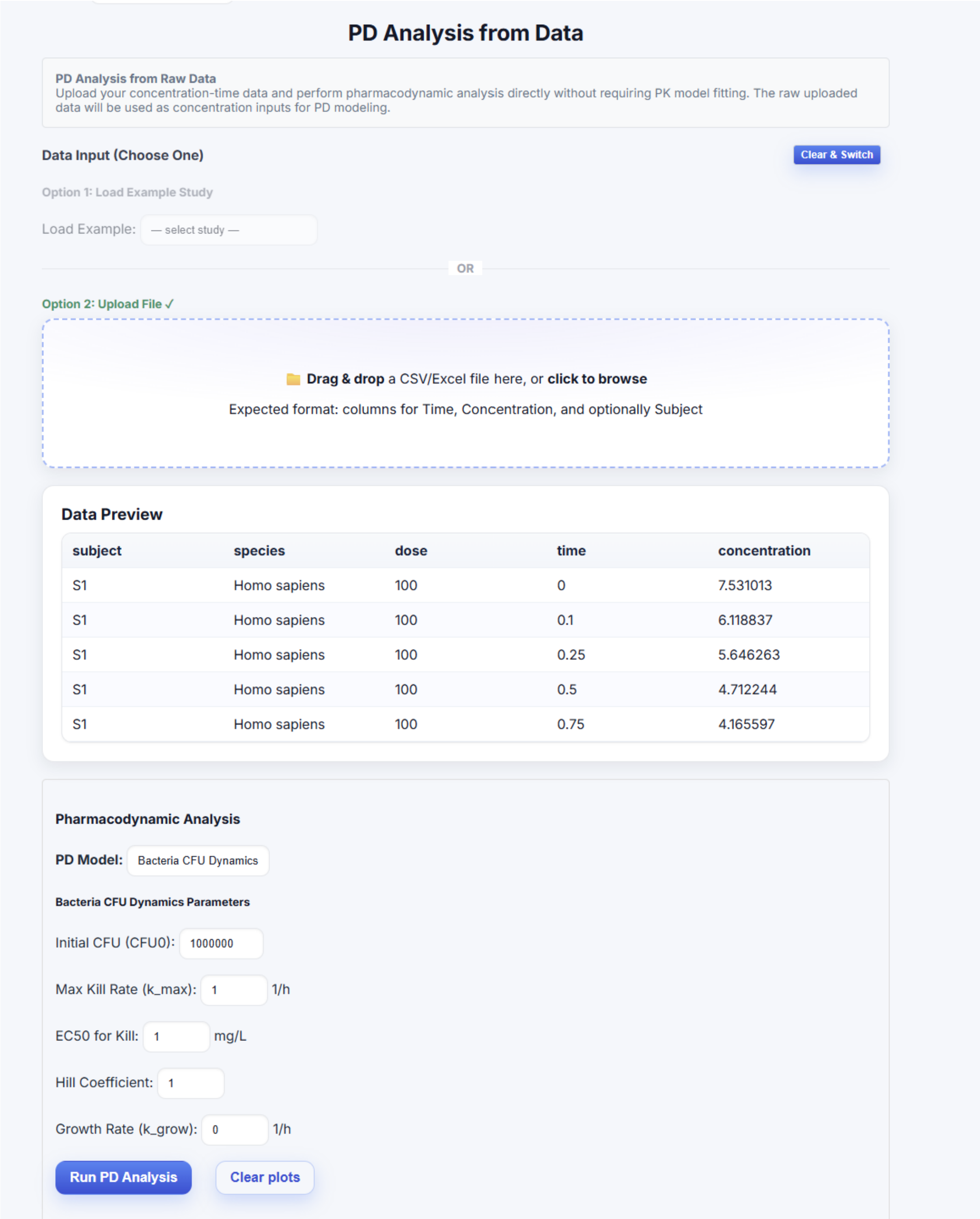

Figure 12: PD Analysis from Data.

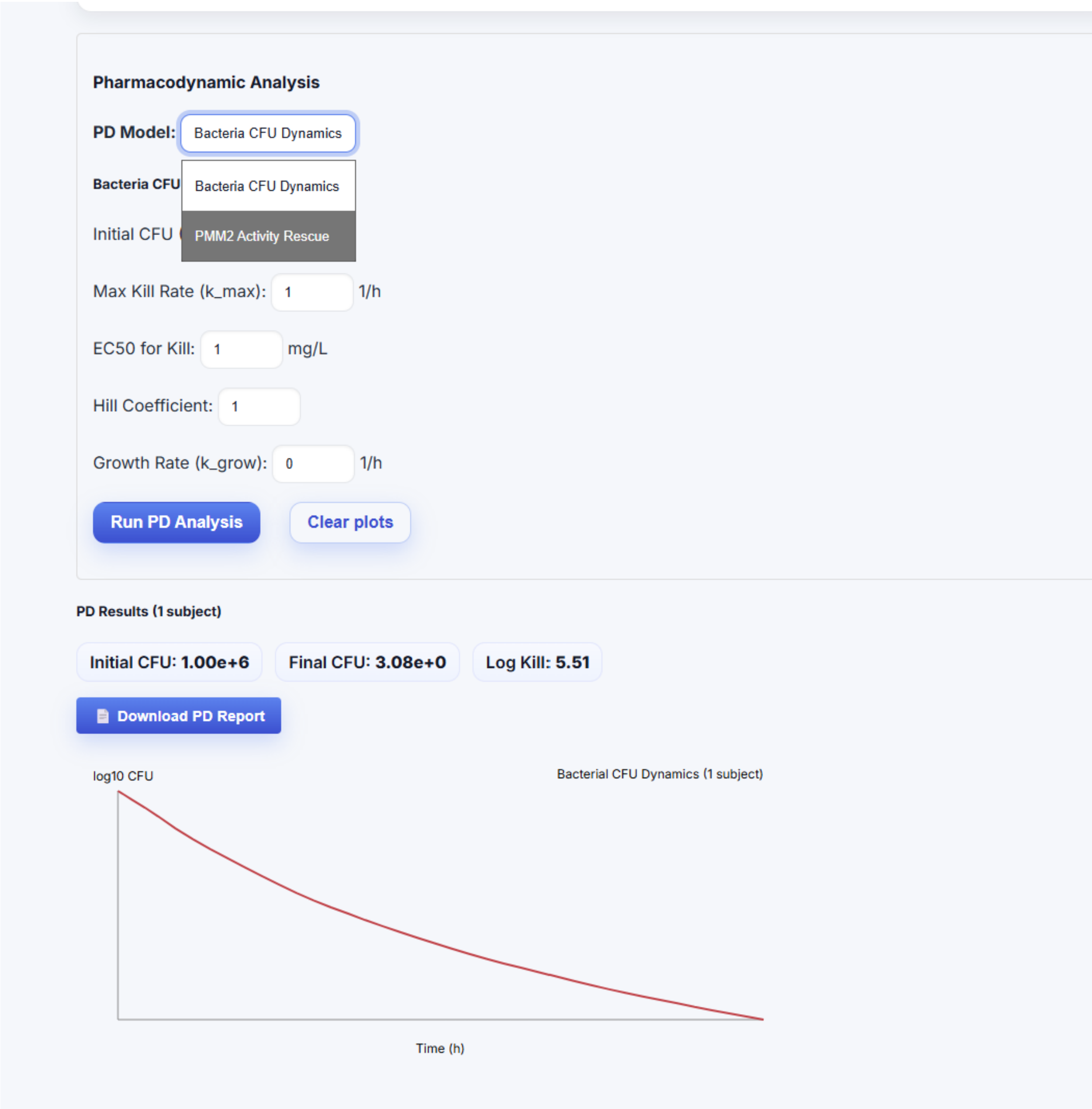

Figure 13: Standalone PD Workbench: Bacteria CFU Dynamics.

15 References

- Internal project docs on datasets, selection/demotion logic, and route formulas (see comment citations at top of this LaTeX file).
- App (hosted): igem.ardabakici.com